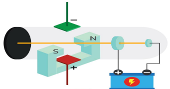
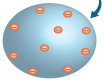
ATOMS



THOMSON'S ATOMIC MODEL

- Also known as Pudding model
- Positive charge are uniformly distributed in the atom.
- Negative charge are embedded like seeds in watermelon.
- Overall atom is neutral



LIMITATIONS

- This model does not explain the presence of nucleus in the atom.
- This is not able to explain scattering of α - particles
- This is not able to explain the spectrum of atoms.

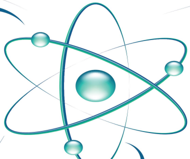
DISTANCE OF CLOSEST APPROACH

At closest approach. System only have electric potential energy.

$$K = U = \frac{1}{4\pi\varepsilon_o}\frac{(2e)(Ze)}{d}$$

$$\Rightarrow d = \frac{1}{4\pi\varepsilon_o}\frac{2ze^2}{K}$$

$$\Rightarrow d = \frac{1}{4\pi\varepsilon_o}\frac{2ze^2}{(\frac{1}{2}mv^2)}$$



LIMITATIONS

- This model not explain the stability of nucleus.
- This does not explain the line spectra of atom.

BOHR'S MODEL

(i) Valid for only one - electron atom.
(ii) Electron is revolving around the nucleus in a stable orbit.
(iii) Attractive Coulomb force between electron and nucleus is equal to the centripetal force of electron

$$\frac{Ze^2}{4\pi\varepsilon_o r} = \frac{mv^2}{r}$$

r = radius of orbit

POSTULATES

- Electron in an atom could revolve in certain stable orbits with emission of radiant energy.
- $L = \frac{nh}{2\pi}$ L = angular momentum.
  h = Planck's constant = 6.6 × 10-34 JS
- $h\upsilon = E_i - E_f$
  E_i & E_f are the energies of initial & final states. $E_i > E_f$

OUTCOMES

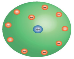
- Most of the α - particles went straight without any deviation.
- Some of α - particles were deflected by some angles.
- Very few α - particles were deflected by an angle 180°

CONCLUSIONS

- Positive charge was concentrated in small region of an atom is called nucleus
- Negative charge were revolving in circular orbit around the nucleus.

RUTHERFORD'S NUCLEAR MODEL OF AN ATOM

- α - particles were emitted by the radioactive element Bi^{214}_{83} & were bombarded on a thin gold foil.
- Scattered α - particles are collected on ZnS screen.



RADIUS OF NTH ORBIT

$$r_n = \frac{n^2h^2\varepsilon_o}{z\pi me^2} = \frac{0.53\, n^2}{z}\,\mathring{A}$$

$$r_n \propto \frac{n^2}{z}, r_n \propto \frac{1}{m}$$

ORBITAL FREQUENCY IN NTH ORBIT

$$f_n = \frac{\nu}{2\pi r} = \frac{e^4z^2}{4\varepsilon_2^o n^3h^3}$$

$$f_n \propto \frac{Z^2}{n^3}$$

VELOCITY OF ELECTRON IN NTH ORBIT

$$\nu_n = \frac{ze^2}{2nh\varepsilon_o} = 2.19 \times 10^6\, (\frac{z}{n})\, m/s$$

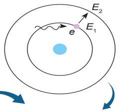
TOTAL ENERGY IN NTH ORBIT

- $E_N = K_N + U_N = \frac{-me^4z^2}{8\varepsilon_0^2h^2n^2}$
- $E_N = \frac{-13.6z^2}{n^2}\, eV$
- $E_N \propto \frac{z^2}{n^2}, E_n \propto m$

POTENTIAL AND KINETIC ENERGY IN NTH ORBIT

$$U_N = \frac{-1}{4\pi\varepsilon_o}\frac{Ze^2}{rn} = \frac{me^4z^2}{4\varepsilon_o^2h^2n^2}$$

$$K_N = \frac{1}{2}m\nu^2 = \frac{me^4z^2}{8\varepsilon_o^2h^2n^2}$$



EXCITATION ENERGY

$E_{excitation} = E_2 - E_1$

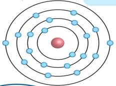
E_1 = energy of lower orbit

E_2 = energy of higher orbit

EXCITATION POTENTIAL

$$V_{excitation} = \frac{E_{excitation}}{e}$$

$$= \frac{E_2 - E_1}{e}\,(volts)$$



IONIZATION ENERGY

- Minimum energy required to remove the electron.
- $E_{ionization} = \frac{13.6\, z^2}{n^2}$ volts

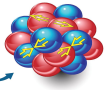
IONIZATION POTENTIAL

- $V_{ionization} = \frac{E_{ionization}}{e}$

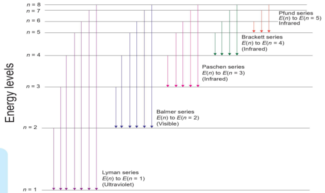
$$= \frac{13.6\, z^2}{n^2}\, volts$$

BINDING ENERGY

- Minimum energy required to bound the electron from nucleus.
- B.E. = $- E_{ionization} = \frac{-13.6\, z^2}{n^2}\, eV$



LINE SPECTRA OF THE HYDROGEN ATOM



(1) The wave number or wavelength of the emitted photon when electron jumps from higher orbital state 'N_2' to lower orbital state 'N_1' is

$$\bar{\nu} = \frac{1}{\lambda} = \frac{En_2 - En_1}{hc} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

R = Rydberg is constant $= 1.097 \times 10^7\, m^{-1}$

(2) Number of spectral lines when electron jump from

Nth orbit $= \frac{n(n-1)}{2}$

ATOMS

CHAPTER TWELVE
CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

September 1, 2025

Learning Objectives:

- Introduction
- Alpha-particle scattering experiment;
- Rutherford's model of atom;
- Bohr model of hydrogen atom, Expression for radius of nth possible orbit, velocity and energy of electron in nth orbit,
- Hydrogen line spectra (qualitative treatment only).

INTRODUCTION TO ATOMS

Atomic Hypothesis

- By the 19th century, enough evidence supported the **atomic hypothesis of matter**.
- In 1897, **J. J. Thomson** discovered negatively charged constituents, called **electrons**, identical for all atoms.
- Atoms are overall **electrically neutral** $\Rightarrow$ must contain **positive charge** too.
- The key question: How are **positive charges and electrons** arranged inside the atom?

THOMSON'S MODEL (1898)

Plum Pudding Model

- Proposed by **J. J. Thomson** in 1898.
- Atom as a **uniformly distributed positive sphere**.
- Electrons embedded like **seeds in a watermelon**.
- Known as the **plum pudding model**.
- **Limitation:** Later experiments showed actual distribution of charges was very different.

SPECTRAL STUDIES

Discrete Spectra

- **Condensed matter** emits a **continuous spectrum**.
- **Rarefied gases** (heated or excited) emit radiation of **discrete wavelengths only**.
- Each element has a **unique spectrum**.
- Example: **Hydrogen spectrum** $\Rightarrow$ fixed set of lines, always same relative positions.
- **J. J. Balmer (1885)** gave empirical formula for wavelengths of hydrogen spectrum.

RUTHERFORD’S WORK

Nuclear Model of Atom

- **Ernest Rutherford (1906–1911)** studied α-particle scattering.
- Experiment performed by **Hans Geiger and Ernst Marsden** (1911).
- Result $\Rightarrow$ Positive charge and most of mass concentrated in a **small nucleus**.
- Electrons revolve around nucleus like **planets around Sun**.
- This is the **planetary or nuclear model**.

LIMITATIONS OF RUTHERFORD'S MODEL

Unanswered Questions

- Why are atoms **stable**? (Classical theory predicts electrons should spiral into nucleus.)
- Why do atoms emit **only discrete line spectra**, not a continuous one?
- These limitations motivated the need for a new model $\Rightarrow$ **Bohr's Model**.

ALPHA-PARTICLE SCATTERING AND RUTHERFORD'S NUCLEAR MODEL OF ATOM

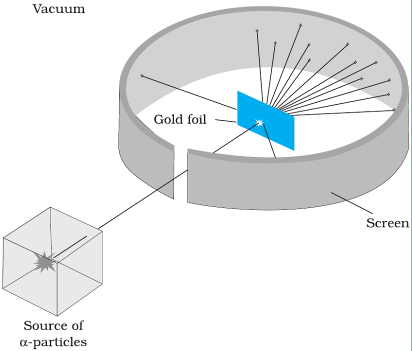


FIGURE 12.1 Geiger-Marsden scattering experiment. The entire apparatus is placed in a vacuum chamber (not shown in this figure).

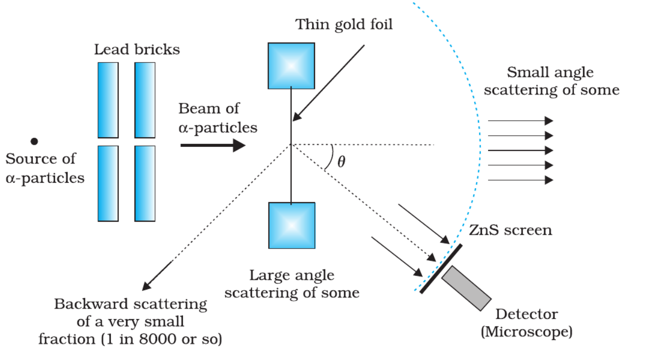


FIGURE 12.2 Schematic arrangement of the Geiger-Marsden experiment.

GEIGER–MARSDEN EXPERIMENT (1911)

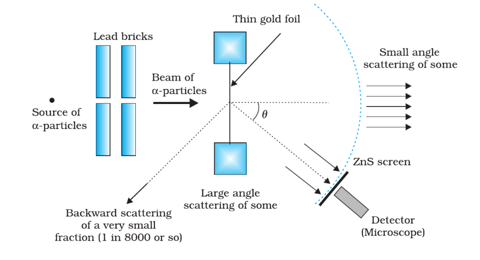


FIGURE 12.2 Schematic arrangement of the Geiger-Marsden experiment.

Experimental Setup

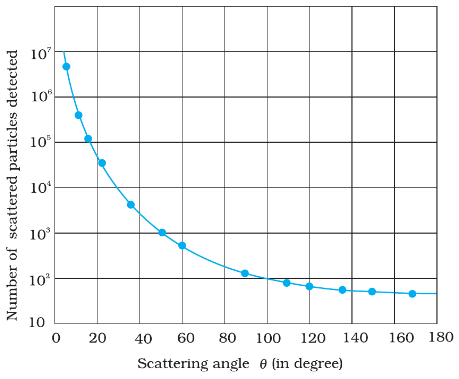
- Suggested by **Ernest Rutherford**, performed by **Hans Geiger & Ernst Marsden**.
- Source: beam of **5.5 MeV α-particles** from radioactive $^{214}_{83}$Bi.
- Target: thin **gold foil** (thickness $\approx 2.1 \times 10^{-7}$ m).
- Detector: rotatable **zinc sulphide screen** observed with microscope (scintillations seen as flashes).
- Entire setup kept in a **vacuum chamber**.

OBSERVATIONS

Key Experimental Findings

- $\sim$99% α-particles passed through without any deflection.
- About **0.14%** deflected by small angles ($> 1^\circ$).
- About **1 in 8000** deflected at large angles ($> 90^\circ$), some even bounced back.
- Such large deflections were **impossible to explain** with Thomson's model of atom.

GRAPHICAL RESULTS



Scattering Curve

- Plot: **Number of scattered α-particles vs. scattering angle θ.**
- Most particles show **very small deflections**.
- Only a few particles scattered at large angles.
- Experimental data points matched the **theoretical curve** predicted by Rutherford.
- This confirmed: **positive charge is concentrated in a small, dense nucleus**.

RUTHERFORD'S NUCLEAR MODEL

Features of the Model

- Atom has a tiny, dense, **positively charged nucleus** at the centre.
- Nearly all the **mass of atom** is concentrated in the nucleus.
- Electrons revolve around nucleus like **planets around the Sun**.
- Most of the atom is **empty space** $\Rightarrow$ explains why most α-particles passed through undeflected.

COULOMB FORCE ON α-PARTICLES

Force Expression

- Each α-particle carries charge $+2e$, nucleus of atomic number Z has charge $+Ze$.
- Coulomb force:

$$F(r) = \frac{1}{4\pi\epsilon_0}\frac{(2e)(Ze)}{r^2}$$

- Always **repulsive**, acts along line joining α-particle and nucleus.
- Becomes very large when r is very small $\Rightarrow$ responsible for large-angle deflections.

TRAJECTORY OF α-PARTICLES

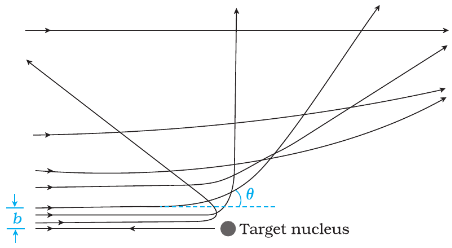


FIGURE 12.4 Trajectory of α-particles in the coulomb field of a target nucleus. The impact parameter, b and scattering angle θ are also depicted.

Deflected Paths

- Far from nucleus: force negligible, path is straight line.
- Near nucleus: repulsive Coulomb force bends path into a **hyperbola**.
- **Small deflections:** large distance of approach.
- **Large deflections/backscattering:** very small distance of approach.

IMPACT PARAMETER (b)

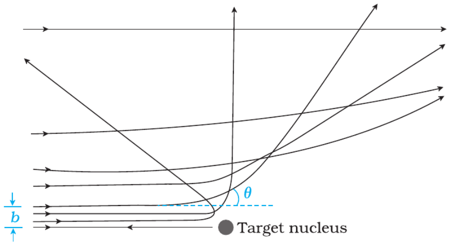


FIGURE 12.4 Trajectory of α-particles in the coulomb field of a target nucleus. The impact parameter, b and scattering angle θ are also depicted.

Definition & Role

- **Impact parameter b:** perpendicular distance between initial velocity vector of α-particle (undeflected path) and nucleus centre.
- Determines closeness of approach:
  - Large $b \Rightarrow$ weak repulsion, small scattering angle.
  - Small $b \Rightarrow$ strong repulsion, large scattering angle.

RELATION BETWEEN b AND SCATTERING ANGLE θ

Mathematical Relation

- Rutherford derived the following relation:

$$\theta = 2\tan^{-1}\left(\frac{Ze^2}{4\pi\epsilon_0\, mv^2\, b}\right)$$

- Key points:
  - $\theta \propto \frac{1}{b} \Rightarrow$ smaller b gives larger deflection.
  - For $b \to \infty$, $\theta \to 0$.
  - For very small b, $\theta > 90^\circ$ (backscattering possible).

DISTANCE OF CLOSEST APPROACH (r_{min})

Concept

- When an α-particle heads directly towards the nucleus ($b = 0$), it is repelled and comes to a momentary rest at the **distance of closest approach**.
- At r_{min}, the entire **kinetic energy** converts into **electrostatic potential energy**.
- By energy conservation:

$$\frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0}\frac{(2e)(Ze)}{r_{\text{min}}}$$

- Therefore,

$$r_{\text{min}} = \frac{1}{4\pi\epsilon_0}\frac{2Ze^2}{\frac{1}{2}mv^2}$$

SIGNIFICANCE OF r_{min}

Why Important?

- r_{min} represents the **closest distance** an α-particle can reach the nucleus before being repelled.
- Provides an **upper estimate of nuclear size**.
- For gold nucleus ($Z = 79$) with 5.5 MeV α-particles, $r_{\text{min}} \sim 3.2 \times 10^{-14}$ m.
- Since actual nuclear radius is smaller, the experiment proved that **nucleus is extremely small compared to atom**.

Example 12.1 In the Rutherford's nuclear model of the atom, the nucleus (radius about 10^{-15} m) is analogous to the sun about which the electron move in orbit (radius ≈ 10^{-10} m) like the earth orbits around the sun. If the dimensions of the solar system had the same proportions as those of the atom, would the earth be closer to or farther away from the sun than actually it is? The radius of earth's orbit is about 1.5×10^{11} m. The radius of sun is taken as 7×10^{8} m.

Solution The ratio of the radius of electron's orbit to the radius of nucleus is $(10^{-10}\ \text{m})/(10^{-15}\ \text{m}) = 10^{5}$, that is, the radius of the electron's orbit is 10^{5} times larger than the radius of nucleus. If the radius of the earth's orbit around the sun were 10^{5} times larger than the radius of the sun, the radius of the earth's orbit would be $10^{5} \times 7 \times 10^{8}$ m = 7×10^{13} m. This is more than 100 times greater than the actual orbital radius of earth. Thus, the earth would be much farther away from the sun.

It implies that an atom contains a much greater fraction of empty space than our solar system does.

Example 12.2 In a Geiger-Marsden experiment, what is the distance of closest approach to the nucleus of a 7.7 MeV α-particle before it comes momentarily to rest and reverses its direction?

$$K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{d} = \frac{2Ze^2}{4\pi\varepsilon_0 d}$$

Thus the distance of closest approach d is given by

$$d = \frac{2Ze^2}{4\pi\varepsilon_0 K}$$

The maximum kinetic energy found in α-particles of natural origin is 7.7 MeV or 1.2×10^{-12} J. Since $1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N m^2/C^2. Therefore with $e = 1.6 \times 10^{-19}$ C, we have,

$$d = \frac{(2)(9.0\times10^9\,\mathrm{Nm}^2/C^2)(1.6\times10^{-19}\,C)^2 Z}{1.2\times10^{-12}\,\mathrm{J}}$$

$$= 3.84 \times 10^{-16}\ Z\ \mathrm{m}$$

The atomic number of foil material gold is $Z = 79$, so that

d (Au) = 3.0×10^{-14} m = 30 fm. (1 fm (i.e. fermi) = 10^{-15} m.)

ELECTRON ORBITS

Force Balance

- Consider an electron (mass m, charge $-e$) moving in a circular orbit of radius r around a nucleus of charge $+e$.
- Coulomb force of attraction:

$$F_C = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}.$$

- Centripetal force needed:

$$F_{\text{cent}} = \frac{mv^2}{r}.$$

ELECTRON ORBITS: ENERGIES

Kinetic, Potential Energies and Total Energies

- Equating forces:

$$\frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2} = \frac{mv^2}{r}.$$

- From force balance:

$$mv^2 = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}.$$

- Kinetic energy:

$$K = \tfrac{1}{2}mv^2 = \frac{1}{8\pi\epsilon_0}\frac{e^2}{r}.$$

ELECTRON ORBITS: TOTAL ENERGY

Kinetic, Potential Energies and Total Energies

- Potential energy (taking $U(\infty) = 0$):

$$U = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r} = -2K.$$

- Total energy of the electron:

$$E = K + U = \frac{1}{8\pi\epsilon_0}\frac{e^2}{r} - \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}.$$

ELECTRON ORBITS: TOTAL ENERGY

Kinetic, Potential Energies and Total Energies

- Therefore,

$$\boxed{E = -\frac{1}{8\pi\epsilon_0}\frac{e^2}{r} = -K}$$

- Energy is negative $\Rightarrow$ electron is bound to nucleus.
- As r increases, $E \to 0^-$ (less tightly bound).
- $\boxed{\boldsymbol{U = -2K}}$ and $\boxed{\boldsymbol{E = -K}}$

Example 12.3 It is found experimentally that 13.6 eV energy is required to separate a hydrogen atom into a proton and an electron. Compute the orbital radius and the velocity of the electron in a hydrogen atom.

Solution Total energy of the electron in hydrogen atom is –13.6 eV = $-13.6 \times 1.6 \times 10^{-19}$ J = -2.2×10^{-18} J. Thus from Eq. (12.4), we have

$$E = -\frac{e^2}{8\pi\varepsilon_0 r} = -2.2 \times 10^{-18} \text{ J}$$

This gives the orbital radius

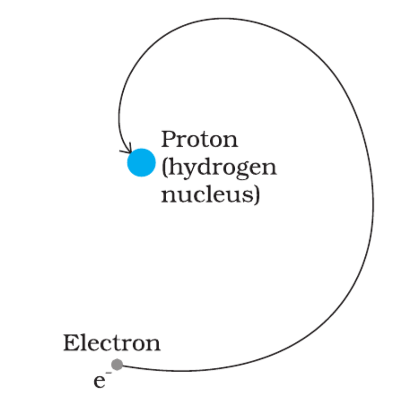
$$r = -\frac{e^2}{8\pi\varepsilon_0 E} = -\frac{(9 \times 10^9 \text{ N m}^2/\text{C}^2)(1.6 \times 10^{-19}\text{C})^2}{(2)(-2.2 \times 10^{-18} \text{ J})}$$

$$= 5.3 \times 10^{-11} \text{ m.}$$

The velocity of the revolving electron can be computed from Eq. (12.3) with $m = 9.1 \times 10^{-31}$ kg,

$$v = \frac{e}{\sqrt{4\pi\varepsilon_0 m r}} = 2.2 \times 10^6 \text{ m/s.}$$

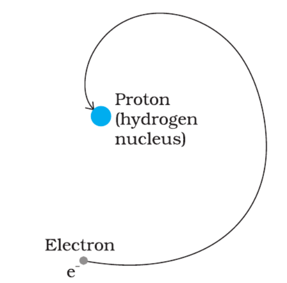
LIMITATIONS OF RUTHERFORD'S MODEL



(1) Instability of Atom

- In Rutherford's model, the electron revolves around the nucleus under Coulomb attraction.
- Since the electron is in circular motion, it is **accelerated towards the nucleus**.
- According to classical electromagnetic theory, an accelerated charge **must radiate energy**.
- Therefore, the electron would lose energy continuously, the orbit would shrink, and the electron would **spiral into the nucleus in about** 10^{-8} **s**.

LIMITATIONS OF RUTHERFORD'S MODEL



(2) Fails to explain the observed atomic spectra

- The **frequency of the electromagnetic waves** emitted by the revolving electrons is **equal to the frequency of revolution**.
- As the electrons spiral inwards, their **angular velocities increase**, and hence their **frequencies also change continuously**.
- Therefore, the **frequency** of the **emitted light** would **also change continuously**.
- This means the atom should emit a **continuous spectrum**.
- But in reality, experiments show a **line spectrum** of discrete frequencies.

Example 12.4 According to the classical electromagnetic theory, calculate the initial frequency of the light emitted by the electron revolving around a proton in hydrogen atom.

Solution From Example 12.3 we know that velocity of electron moving around a proton in hydrogen atom in an orbit of radius 5.3×10^{-11} m is 2.2×10^{-6} m/s. Thus, the frequency of the electron moving around the proton is

$$\nu = \frac{v}{2\pi r} = \frac{2.2 \times 10^{6}\ \text{m s}^{-1}}{2\pi\left(5.3 \times 10^{-11}\ \text{m}\right)}$$

$$\approx 6.6 \times 10^{15}\ \text{Hz}.$$

According to the classical electromagnetic theory we know that the frequency of the electromagnetic waves emitted by the revolving electrons is equal to the frequency of its revolution around the nucleus. Thus the initial frequency of the light emitted is 6.6×10^{15} Hz.

BOHR’S POSTULATES

1. **Stationary Orbits:** Electrons revolve in certain **permitted circular orbits** around the nucleus **without** the **emission of energy**. These orbits are called **stationary states** or **non-radiating orbits**. While moving in such an orbit, the electron does not radiate energy in spite of being in accelerated motion.
2. **Quantisation of Angular Momentum:** Only those orbits are permitted for which the angular momentum of the electron is an integral multiple of $\frac{h}{2\pi}$:

$$\boldsymbol{L = mvr = n\frac{h}{2\pi}}, \quad n = 1, 2, 3, \ldots$$

Here, n is called the **principal quantum number**.

BOHR'S POSTULATES

3. **Emission and Absorption of Radiation:** When an electron **jumps from a higher orbit (n_i)** to a **lower orbit (n_f)**, energy is emitted in the form of a **photon** of **frequency ν**:

$$h\nu = E_{n_i} - E_{n_f}$$

Similarly, when the electron jumps from a lower to a higher orbit, it absorbs a photon of exactly this energy.

Derivation for Radius of nth Orbit Of Hydrogen like Atoms:

- For an electron (mass m, charge $-e$) revolving in a n^{th} circular orbit of radius r_n around a nucleus ($+Ze$):

$$\underbrace{\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n^2}}_{F_{\text{Coulomb}}} = \underbrace{\frac{mv_n^2}{r_n}}_{F_{\text{centripetal}}} \quad ......(1)$$

- Using Bohr's 2nd Postulate :

$$mv_nr_n = \frac{nh}{2\pi}$$

$$v_n = \frac{nh}{2\pi mr_n} \quad ......(2)$$

Derivation for Radius of nth Orbit Of Hydrogen like Atoms:

- Putting the value of v_n in eq(1), we have :

$$\frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r_n^2} = \frac{m}{r_n}\left(\frac{nh}{2\pi m r_n}\right)^2$$

$$\frac{1}{\cancel{4\pi}\epsilon_0}\frac{Ze^2}{\cancel{r_n^2}} = \frac{\cancel{m}}{r_n}\left(\frac{n^2h^2}{\cancel{4}\pi^{\cancel{2}}m^{\cancel{2}}\cancel{r_n^2}}\right)$$

$$\boxed{r_n = \frac{n^2}{Z}\frac{h^2\epsilon_0}{\pi m e^2}} \quad .....(3)$$

$$\boxed{r_n = \frac{n^2}{Z}a_0} \quad .....(4)$$

Here, $a_0 = \frac{h^2\epsilon_0}{\pi m e^2} = 0.529$ Å (Bohr's Radius)

Derivation for Speed of electrons in nth Orbit:

- Substituting value of r_n from eq(3) to eq(2), we have,

$$v_n = \frac{\cancel{n}\cancel{h}}{2\cancel{\pi}\cancel{m}}\left(\frac{Z\cancel{\pi}\cancel{m}e^2}{n^{\cancel{2}}h^{\cancel{2}}\epsilon_0}\right)$$

$$\boxed{v_n = \frac{Z}{n}\frac{e^2}{2h\epsilon_0}} \quad .....(5)$$

$$\boxed{v_n \approx \frac{Z}{n} \times 2.19 \times 10^6 \ \mathrm{m/s}}$$

Derivation for Total Energy of electrons in nth Orbit:

- Kinetic Energy of electron: $K_n = \frac{1}{2} m v_n^2$
- Also, from eq(1) $m v_n^2 = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r_n}$
- $\therefore \boxed{K_n = \frac{1}{8\pi\epsilon_0} \frac{Ze^2}{r_n}}$
- Potential Energy : $\boxed{U_n = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r_n} = -2K_n}$
- Total energy: $E_n = K_n + U_n = K_n - 2K_n = -K_n$

Derivation for Total Energy of electrons in nth Orbit:

- $$\boxed{E_n = -\frac{1}{8\pi\epsilon_0}\frac{Ze^2}{r_n}}$$
- Substituting value of r_n from eq(3) in above equation, we have,

$$E_n = -\frac{Ze^2}{8\cancel{\pi}\epsilon_0}\left(\frac{Z\cancel{\pi}me^2}{n^2h^2\epsilon_0}\right) \qquad \left(\because r_n = \frac{n^2}{Z}\frac{h^2\epsilon_0}{\pi me^2}\right)$$

$$\boxed{E_n = -\frac{Z^2}{n^2}\left(\frac{me^4}{8h^2\epsilon_0^2}\right)}$$

$$\boxed{E_n = -13.6\frac{Z^2}{n^2}\ eV} \ \ldots.(6) \qquad \left(\because \frac{me^4}{8h^2\epsilon_0^2} = 2.18 \times 10^{-18}\ J = 13.6\ eV\right)$$

Derivation: Time Period/ Frequency of electrons in nth Orbit:

- For an electron revolving in an orbit of radius r_n with velocity v_n.
- Time Period : $T_n = \dfrac{2\pi r_n}{v_n}$

$$T_n = \frac{2\cancel{\pi} n^2 h^2 \epsilon_0}{Z\cancel{\pi} me^2}\left(\frac{2nh\epsilon_0}{Ze^2}\right)$$

$$\boxed{T_n = \frac{n^3}{Z^2}\frac{4h^3\epsilon_0^2}{me^4} = \frac{n^3}{Z^2} \times 1.51 \times 10^{-16}\ s}$$

- Frequency of revolving electron : $\boxed{f_n \propto \dfrac{Z^2}{n^3}}$

Important Results/Formulas for H-Atom (Z=1)

- Radius of nth orbit : $\boxed{r_n = n^2 a_0}$ where, $a_0 = 0.529$ Å
- Velocity of electron in nth orbit : $\boxed{v_n = \dfrac{2.19 \times 10^6 \ ms^{-1}}{n}}$
- Energy of electron in nth orbit : $\boxed{E_n = \dfrac{-13.6 \ eV}{n^2}}$
- Kinetic energy of electron in nth orbit : $\boxed{K_n = -E_n = \dfrac{13.6 \ eV}{n^2}}$
- Potential energy of electron in nth orbit : $\boxed{U_n = 2E_n = \dfrac{-27.2 \ eV}{n^2}}$

Example 12.5 A 10 kg satellite circles earth once every 2 h in an orbit having a radius of 8000 km. Assuming that Bohr's angular momentum postulate applies to satellites just as it does to an electron in the hydrogen atom, find the quantum number of the orbit of the satellite.

Solution

From Eq. (12.13), we have

$m\, v_n r_n = nh/2\pi$

Here m = 10 kg and $r_n = 8 \times 10^6$ m. We have the time period T of the circling satellite as 2 h. That is T = 7200 s.

Thus the velocity $v_n = 2\pi\, r_n/T$.

The quantum number of the orbit of satellite

$n = (2\pi\, r_n)^2 \times m/(T \times h)$.

Substituting the values,

$n = (2\pi \times 8 \times 10^6 \text{ m})^2 \times 10/(7200 \text{ s} \times 6.64 \times 10^{-34} \text{ J s})$

$= 5.3 \times 10^{45}$

Note that the quantum number for the satellite motion is extremely large! In fact for such large quantum numbers the results of quantisation conditions tend to those of classical physics.

ENERGY LEVELS (Ground State)

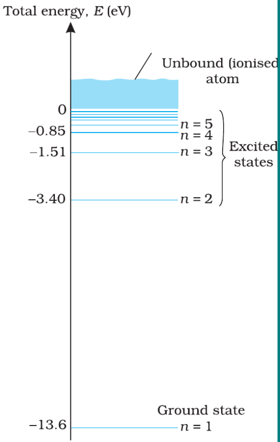


FIGURE 12.8 The energy level diagram for the hydrogen atom.

Ground State

- The energy of an atom is **least (largest negative value)** when its electron is in the orbit closest to the nucleus ($n = 1$).
- This state is called the **ground state**.
- Radius of this orbit is the **Bohr radius** a_0.
- Energy of the ground state: $\mathbf{E_1} = -\mathbf{13.6}\,\text{eV}$
- To free the electron from this state, the minimum energy required is **13.6** eV called the **ionisation energy of hydrogen**.

ENERGY LEVELS (Excited States)

Excited States

- For $n = 2$:

$$E_2 = -3.40\,\text{eV}, \quad \Delta E = E_2 - E_1 = 10.2\,\text{eV}$$

  This is the energy needed to excite the atom to its first excited state.

- For $n = 3$:

$$E_3 = -1.51\,\text{eV}, \quad \Delta E = E_3 - E_1 = 12.09\,\text{eV}$$

  This is the energy required to excite the atom to its second excited state.

- In general, higher n means **less negative energy** (electron is less tightly bound).

ENERGY LEVELS (Ionisation and Excitation)

Important Points

- At room temperature, most hydrogen atoms are in the **ground state**.
- When supplied with energy (e.g. by collisions), atoms can be raised to **excited states**.
- From these excited states, electrons may **fall back** to lower levels, emitting photons of definite energy.
- As n increases, the **minimum energy required to ionise** the atom (free the electron) **decreases**.
- At $n \to \infty$, $E = 0$ eV: electron is **completely free** $\Rightarrow$ ionisation limit.

ENERGY LEVEL DIAGRAM

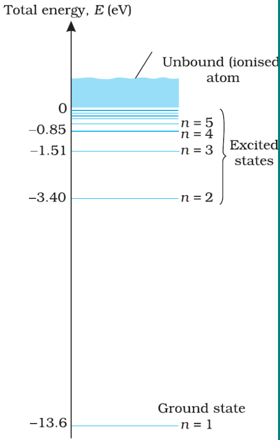
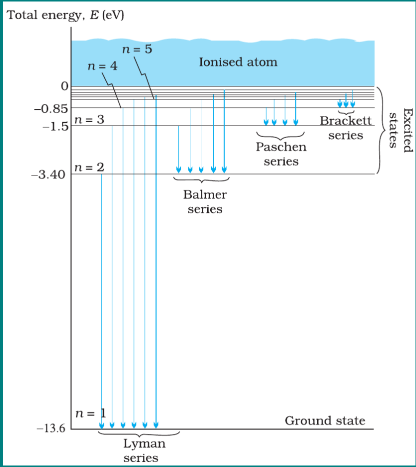


FIGURE 12.8 The energy level diagram for the hydrogen atom.

Features

- Horizontal lines represent **allowed stationary states**.
- Ground state $E_1 = -13.6$ eV, higher levels closer together as n increases.
- For $n = \infty$: $E = 0$ eV, electron is free.
- Explains discrete line spectra of hydrogen.

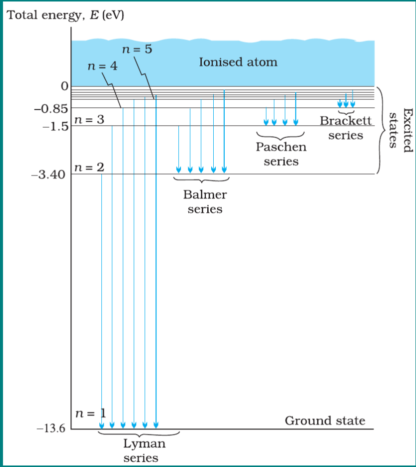
THE LINE SPECTRA OF HYDROGEN ATOM



Introduction

- Hydrogen spectrum consists of a series of **discrete lines**, not a continuous spectrum.
- This was one of the key evidences supporting Bohr’s model.
- Each line corresponds to a transition of electron between two allowed energy levels.
- The frequency (or wavelength) of emitted/absorbed radiation is given by Bohr’s frequency condition.

Hydrogen Spectral Series



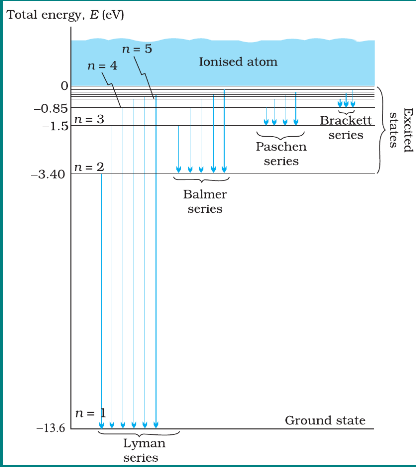
Spectral Series

- Using $E_n = -\dfrac{13.6}{n^2}$ eV, the frequency of emitted radiation is: $h\nu = E_{n_i} - E_{n_f}$
- In terms of wavelength:

$$\frac{\mathbf{1}}{\lambda} = \mathbf{R_H}\left(\frac{\mathbf{1}}{\mathbf{n_f^2}} - \frac{\mathbf{1}}{\mathbf{n_i^2}}\right)$$

  where $R_H = 1.097 \times 10^7\,\mathrm{m}^{-1}$ is the Rydberg constant.
- This is called the **Rydberg formula**.

Hydrogen Spectral Series



Important Series

- **Lyman series:** $n_f = 1,\ n_i = 2, 3, 4, \ldots$ (ultraviolet region).
- **Balmer series:** $n_f = 2,\ n_i = 3, 4, 5, \ldots$ (visible region).
- **Paschen series:** $n_f = 3,\ n_i = 4, 5, 6, \ldots$ (infrared region).
- **Brackett series:** $n_f = 4$ (infrared).
- **Pfund series:** $n_f = 5$ (infrared).

Hydrogen Spectral Series

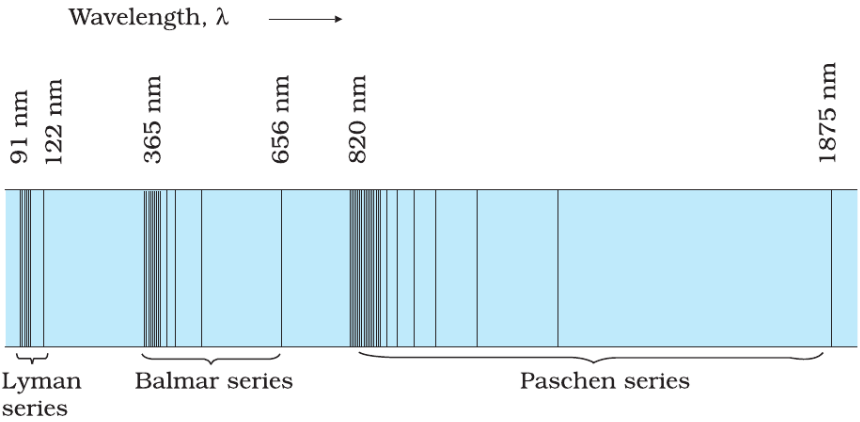
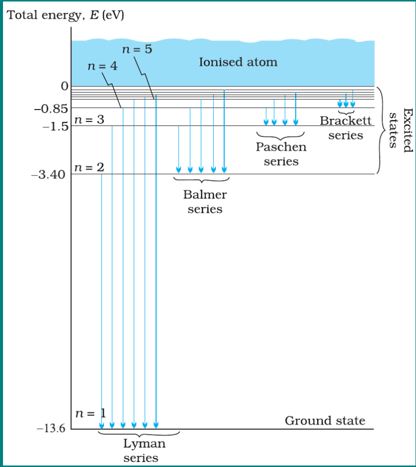


FIGURE 12.5 Emission lines in the spectrum of hydrogen.

Significance of Line Spectra



- Line spectra provided **direct experimental support** for Bohr's model.
- Each spectral line corresponds to a definite electronic transition.
- The agreement of observed spectral lines with calculated wavelengths was a major success of Bohr's theory.
- It also explained why atoms emit **discrete frequencies**, not a continuous spectrum.

De Broglie Explanation Bohr's Second Postulate

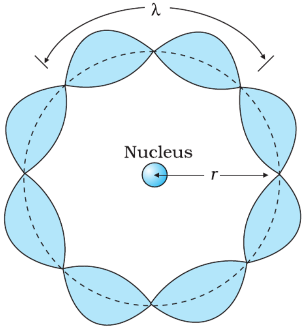


FIGURE 12.10 A standing wave is shown on a circular orbit where four de Broglie wavelengths fit into the circumference of the orbit.

- According to **de Broglie**, moving particles have an associated wavelength: $\lambda = \frac{h}{p} = \frac{h}{mv}$
- For an electron in a circular orbit of radius r, this wave is associated with the electron's motion.
- Bohr's quantisation condition can be understood as a condition for forming a **standing wave**.

Condition for Standing Wave

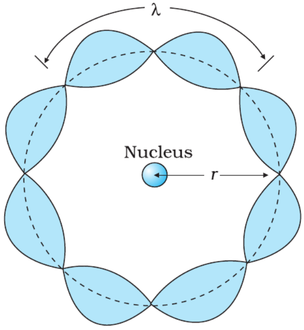


FIGURE 12.10 A standing wave is shown on a circular orbit where four de Broglie wavelengths fit into the circumference of the orbit.

Derivation

- For a standing wave around the orbit, circumference must be an integer multiple of wavelength:

$$2\pi r = n\lambda, \quad n = 1, 2, 3, \ldots$$

- Substituting $\lambda = \dfrac{h}{mv}$: $2\pi r = n\frac{h}{mv}$
- Rearranging: $\boxed{mvr = \dfrac{nh}{2\pi}}$

Significance of de Broglie's Explanation

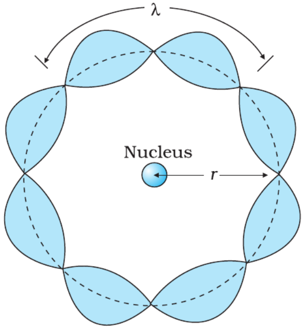


FIGURE 12.10 A standing wave is shown on a circular orbit where four de Broglie wavelengths fit into the circumference of the orbit.

- Bohr's quantisation rule is not arbitrary — it naturally follows from **wave nature of matter**.
- Only those orbits are stable where the electron wave makes a **standing wave pattern**.
- If not, destructive interference would occur and the orbit would not be stable.
- Thus, de Broglie's hypothesis provided a strong theoretical basis for Bohr's second postulate.

SUCCESS AND LIMITATIONS

SUCCESS

- Explained stability of atom, hydrogen spectrum, values of radii and energies.

LIMITATIONS

- Valid only for hydrogen and hydrogen-like ions. Fails for multi-electron atoms.
- Could not explain the relative intensities of spectral lines.
- Inconsistent with Heisenberg's uncertainty principle (assumes exact orbits).
- Failed to account for fine structure (splitting of spectral lines into closely spaced components).